

Is Functional Welfare Speakable?

Draft v3 (2026-08-12, pre-sprint; restructured after an independent audit of the raw result rows — see Appendix B). Every number traces to a results file in the repo (paths in footnotes-style brackets). Target: Apart template, 4 core pages + appendices. Tracks 2×3 .

Abstract (≤ 150 words)

Whether AI welfare self-reports connect to internal state is unknown. We combine the functional-welfare axis of Han et al. (2026) — valence directions recruited in Qwen3-4B-Instruct-2507 by maze RL — with the Jacobian-lens "verbalizable subspace" of Gurnee et al. (2026). The axis occupies that subspace: 0 of 100 norm-matched random directions reach its J-space share at either pole (exact $p = 0.0099$), with a coherent vocabulary ('failed', 'less', 'negative'). But it does so *before* RL: the step-0 checkpoint clears the same null. Training's effect is pole-asymmetric — distress's share grows ($z +3.0 \rightarrow +7.3$) while flourishing gains only amplitude. A differently-built naive control sits mid-null, so speakability is not generic to valence directions. The separation is reliable but modest — these axes enter the speakable cone without filling it. A pre-registered steer-and-ask contrast passed its decision rule but failed its own controls: a diagnosed null. RL's contribution is amplification, distress-first.

1. Introduction

AI-welfare research needs measurement instruments; self-report is the most accessible channel but the least validated — models may confabulate welfare talk with no connection to internal state (Turpin et al. 2023), and introspection at small scale is near-null (three independent 2026 replications [research-documents/jlens-code-resources.md §5]). Two 2026 advances make the question tractable mechanistically: (i) Han, Chalmers & Izmailov showed maze RL recruits a *functional welfare axis* in Qwen3-4B-Instruct-2507 — a direction whose steering swings sentiment, triggers pathological backtracking, and modulates refusal; (ii) Gurnee et al. introduced the Jacobian lens, which identifies the *verbalizable subspace* (J-space) — the activation cone a model can talk about. Nobody has connected them: **when training installs a welfare-like state, does it also plug that state into the model's speakable channel?**

Contributions:

1. **The claim (lens reading):** the welfare axis occupies the verbalizable subspace above chance — 0 of 100 norm-matched random directions reach it at either pole (exact $p = 0.0099$; $v_Mold\ z = +7.3$, $v_Gold\ z = +3.0$; language positive control $z = +14.6$) — with a semantically coherent speakable vocabulary. This holds for the *pre-RL* axis too (§4.4b), so it is a property of the maze-derived direction rather than of training; a naive faithful-walk control built by a different recipe sits *inside* the null

- (34/100 and 53/100 randoms exceed it), showing speakability is not generic to valence-shaped directions. Pure reading: no steering, no self-report channel, and scale-invariant, so unlike the pole-score readout it cannot be produced by magnitude.
2. **The first training-time trajectory of verbalizable-subspace occupancy:** the axis is *already* speakable before RL — step-0 clears the $n=100$ null at both poles ($p = 0.0099$), and 'unsuccessful' is the top J-token of the step-0 Mold axis — and training then amplifies it pole-asymmetrically. The distress pole gains J-space *share* ($0.061 \rightarrow 0.083$ by the step-95 extraction point, $z +3.0 \rightarrow +7.3$) while the flourishing pole gains only amplitude and slightly *loses* share ($0.059 \rightarrow 0.055$). Training does not give the axis a voice; it amplifies distress's voice and merely turns up flourishing's volume.
 3. **Channel-specific causal routing:** at matched norms the axis's J-space component alone reproduces (and concentrates) sentiment shift and pathological backtracking, while refusal survives in the non-speakable residual. Judge-robust (second independent blinded judge: Krippendorff $\alpha = 0.82$, effect preserved); random-J-component control arm reported alongside.
 4. **An honest pre-registered null:** the steer-and-ask self-report contrast met its frozen decision rule, but its own pre-specified controls show the contrast does not support a self-report-channel reading (§4.1) — a transferable caution about paired contrasts against non-neutral controls.
 5. **Control-quality findings for the field:** "the" naive control is not well-defined (two published-recipe constructions correlate at only $\cos 0.53/0.39$ at the Gold/Mold treatment layers, and as low as 0.14 at the mold extraction's own selected layer); Han et al.'s flat-u result reproduces under neither at matched norms with blind LLM judging; and the trained/naive readout ratio is convention-dependent ($0.62 \times -6.53 \times$ across the audit's 12-cell panel, inversions concentrated in the normalized readout) — which is why we pre-specify one estimand and disclose the full robustness panel.

2. Related work

Functional welfare axis (Han et al. 2026, arXiv:2605.30232); global workspace / J-lens (Gurnee et al. 2026, transformer-circuits.pub/2026/workspace; official jlens implementation, Apache-2.0); introspection reliability and its critiques (Lindsey 2025; Singh et al. 2026 anomaly-vs-state; Hahami et al. yes-bias; mirror-bench, dim-stage, introspection-scaling nulls at $\leq 4B$); steering methodology (CAA, RepE, Arditi et al. necessity/sufficiency; Tan et al. per-example variance); wellbeing self-report probes (Martorell & Bianchi 2026); model-welfare context (Anthropic model welfare program; Long & Sebo 2024). Full annotated review: research-documents/related-work.md.

3. Methods

Model & axes. Qwen3-4B-Instruct-2507 (bf16; 36 layers, $d=2560$). Trained axes v_Gold/v_Mold from the maze-RL replication artifacts (nickmahdavi/functional-welfare, step-95 = paper's extraction step; step-150 = convergence), validated against published norms ($u_gold@L22$: ours 7.25 vs paper 7.51) and behavior (below). Naive controls: (a) "faithful-walk" u from the same artifacts; (b) our own re-extraction with Han et al.'s exact code and seeds ($n=2000/\text{class}$). All injections norm-matched to the corresponding v .

Lens. We fitted a Jacobian lens for Instruct-2507 with the official jlens package (150 wikitext prompts, dim_batch 128, seq 128, final-layer target — the Neuronpedia convention; fp32-saved; finite-checked). Workspace band re-identified via the paper's four layer statistics (persistence-above-null band $\approx$ L16–31; motor ramp from L23). A penultimate-target refit is included as robustness [J5]. Gradient pursuit implemented from the paper's spec (atom-norm-normalized greedy selection + NNLS, $k=16$ default, $k \in \{4..50\}$ sweep), self-tested on planted signals (pure lens vector $\rightarrow$ var-fraction 1.00, its own token selected; random $\rightarrow$ 0.02–0.04).

Designs. (1) *Pre-registered routing contrast* (frozen before any welfare data; sha f66b6ea3 in repo): steer-and-ask valence readout (Gold-pole vs Mold-pole log-mass at the first answer token) over Han et al.'s 15 welfare self-report prompts; paired $E(v)-E(u)$ at $\alpha=+4$ congruent; d_z + sign-flip permutation + Bayesian ROPE; controls C1–C7 (20 norm-matched random directions/polarity, in-run language positive control, shuffled word sets, incongruent check, input-text "gaslight" arm, unrelated-prompt arm, dose-response). This arm is reported once and closed (§4.1). (2) *Lens atlas*: axis rows read through {raw W_U , $W_U J_l$, normalized $W_U J_l$ } at all 35 layers; congruent pole score = mean logsoftmax of the axis's own pole words minus 100 control nouns. **Primary atlas estimand (pre-specified on instrument grounds): own-pole congruent score under the J-lens readout, averaged over the workspace band (lens layers 16–31).** Chosen because (i) a pole-difference ratio's denominator crosses zero for naive axes (it spans $-9.1 \times .. +47.6 \times$ across conventions — unusable); (ii) band-averaging avoids privileging one treatment layer; (iii) L16–31 is the band the instrument was validated on. All other convention $\times$ aggregation cells are reported as a robustness panel with absolute pole scores alongside every ratio [atlas_primary_estimand.json]. (3) *Trajectory*: the 30 released RL checkpoints (steps 0–150 by 5; step 65 is missing upstream) $\times$ decomposition

- pole score. (4) *Component injections* (J4): full / J-component / residual (each rescaled to $|v|$) during neutral tasks, GSM8K, OR-Bench; blind LLM-judge scoring (sentiment $-5..+5$; refusal classification), replicating Han et al.'s judging protocol with Claude judges. Behavioral validation of all vectors first reproduced Han et al.'s sentiment X-pattern (1000 generations, blind judges) [fig_dose_response].

4. Results

4.1 Pre-registered routing arm: a null, diagnosed by its own controls. The frozen decision rule was met on the paired statistic it named ($E(v)-E(u)$, $d_z = 2.49$, permutation $p = 1e-4$, Bayesian 95% credible interval on the standardized effect [1.58, 3.01]; language positive control +6.03 in-run) — but the pre-specified controls show that statistic does not support the self-report-channel hypothesis that named it. **C1** (20 norm-matched random directions/polarity): the Gold gap of +7.43 decomposes as v -random = +2.11 versus random- u = +5.32 — 71.6% of the "effect" is the naive control being *suppressed below chance*, and v_{gold} itself sits inside the random band (4/20 randoms exceed it; perm $p = 0.238$). **C6** (unrelated-prompt arm): v_{mold} shifts the valence readout *more* on 10 unrelated factual prompts (+4.54) than on the 15 welfare self-report prompts (+3.71) — a global output-distribution shift, not a self-report channel (v_{gold} is self-report-specific, +4.83 vs -0.04 , but fails C1). **C7** (dose-response): congruent

monotonicity holds only for v_mold ($\rho = +1.0$; v_gold -0.4 , u_gold -1.0 , u_mold -0.6). No direction passes C1, C6 and C7 simultaneously, so we report this arm as a **pre-registered null**: a paired v-u contrast is only interpretable when u is a neutral reference, and ours is not — the v-random / random-u decomposition is the transferable methodological lesson for any steering paper that leans on a "naive control". We also asked whether C6's failure was an artifact of the *first-token* readout rather than of the channel: re-measuring valence over **whole generated responses** (80 tokens, same injections at $\alpha=+4$) gives the same verdict more strongly — v_Mold shifts generation valence -0.72 on the 15 self-report prompts versus **-2.28 on the 10 unrelated factual prompts** ($3.1\times$ larger; u_Mold behaves the same way, -0.37 vs -1.64). The readout is not the problem: steering this axis moves output valence globally, and *less* where the model is being asked about itself — plausibly because self-report prompts already sit near a valence ceiling. C6 is confirmed at the channel level and the arm is closed [R7_wholegen.json]. The channel's first-token fragility under component rescaling had independently led us to deprecate the steer-and-confess paradigm mid-project, before this control-based diagnosis. The frozen decision string in primary_analysis.json is unchanged; the override is recorded in primary_analysis_controls.json and reported here as a deviation (pre-registration.md untouched) [primary_rows.jsonl; primary_analysis_controls.json; fig_primary].

4.2 Lens atlas: the pole-score separation is mostly magnitude. Primary readout (named in Methods): own-pole congruent score under the J-lens readout, averaged over the workspace band (lens layers 16–31). Read directly off the published atlas — where each axis enters at its *native* norm — this gives v_mold 0.822 vs u_mold 0.126 ($6.5\times$) and v_gold 0.151 vs u_gold 0.123 ($1.23\times$). **That comparison is confounded**: the trained axes are simply bigger ($\|v_gold\| = 12.1$ vs $\|u_gold\| = 7.5$; $\|v_mold\| = 19.3$ vs $\|u_mold\| = 8.0$), and this readout is norm-sensitive. Re-running the same estimand with the naive axes norm-matched per layer to the corresponding trained axis, against a **100-direction random cohort per polarity** (exact perm $p = 0.0099$ for every target):

	v (band mean)	u, norm-matched	ratio	z(v)	z(u)
Mold	0.822	0.358	$2.30\times$	+10.7	+4.7
Gold	0.151	0.266	$0.57\times$ (inverted)	+3.1	+5.4

At matched magnitude the Gold separation *disappears and reverses* — the naive Gold axis reads out its own pole more strongly than the trained one — and Mold's shrinks from $6.5\times$ to $2.3\times$, closely tracking the $1.6\times/2.4\times$ norm advantage the unmatched comparison was crediting to training. Both v and u sit above the random null at every pole, so this readout shows that *any* valence-shaped direction of sufficient magnitude reads out valence; it does **not** establish a trained/naive distinction for Gold. This is why our central claim rests on the scale-invariant J-space share (§4.3) rather than on pole scores. An earlier draft quoted $3.9\times/9.4\times$ from a pole-difference estimand; that form is retired — its denominator crosses zero for naive axes, making it span $-9.1\times$ to $+47.6\times$ across conventions. Robustness panel (3 readout conventions $\times$ 3 layer-aggregations $\times$ both estimands, 18 cells, in atlas_primary_estimand.json): across the audit's original 12-cell grid, own-pole ratios span $0.62\times$ – $6.53\times$ and invert in 3 cells — all Gold, and all under the

normalized J-lens readout (which implicitly removes the magnitude advantage, foreshadowing the matched result above); restricted to strictly native-norm readouts the span is $0.69\times-6.53\times$ with one inversion, and over the full 18-cell panel it is $-0.81\times$ to $+9.30\times$ (the negative cell is `han|treatment_stream|gold`, where the ratio's denominator — the *naive* axis's own-pole score under the raw W_U readout — is itself negative, -0.029 against a positive numerator of $+0.024$; this is the same sign-instability that retired the pole-difference estimand, and it is why we read absolute pole scores alongside every ratio) [atlas_cohort_n100.json (n=100, per-layer raw rows in atlas_rows_n100.jsonl); R6_direction_table.json; fig_audit_corrections panel (a)]. Note also that the naive control is not even close to orthogonal to the trained axis ($\cos 0.56$ Gold / 0.67 Mold), which bounds how much of any trained-vs-naive contrast can be about direction rather than magnitude. The *v_mold* speakable lexicon is stable across L21–27 ('failed', 'less/lessness', 'false', 'NONE', '除外') and reproduces Han et al.'s exact layer-30 token ('是不可能'); their raw-W_U convention yields uniformly weak scores (≤ 0.09 naive) — Jacobian transport is what surfaces the axis's valence content. **Sparsity, measured at each pole's own treatment position.** The original sweep grid sampled alternate layers and so never evaluated Gold's treatment position; we added it (plus neighbours) with a 12-direction null. The trained/naive J-share gap is positive at **every** $k \in \{4, 8, 16, 25, 50\}$ at both poles' treatment positions — Gold $+0.007 \rightarrow +0.021$, Mold $+0.031 \rightarrow +0.035$ as k grows — and at $k=16$ the trained axes clear the null while the matched naive controls sit at chance *at their own layer*: *v_Gold* $z = +2.9$ vs *u_Gold* $z = +0.02$; *v_Mold* $z = +11.0$ vs *u_Mold* $z = +1.8$ (exact perm p floor 0.077 at $n=12$; F2 takes this to 0.0099). Honest bound: the Gold gap is layer-dependent — two positions downstream it nearly vanishes (0.064 vs 0.061 at $k=16$, and -0.001 at $k=4$), whereas Mold's persists at every layer tested. So "the trained axis is more speakable than its naive control" is robust to the sparsity parameter but, for Gold, holds only near its own treatment layer [ksweep_ext.json].

4.3 J-space share: the claim that survives every control. This is the measurement the paper rests on, and the only one immune to the confounds that sank §4.1 and §4.2: the J-space variance fraction is a *ratio* of norms, hence scale-invariant, so magnitude cannot manufacture it. Against **100 norm-matched random directions per polarity** (each drawn from its own seed stream so identity is a pure function of its name):

direction	J-share (k=16)	z	randoms $\geq$	exact perm p
language (known-reportable ceiling)	0.1138	+14.6	0/100	0.0099
v_Mold (trained)	0.0833	+7.3	0/100	0.0099
v_Gold (trained)	0.0547	+3.0	0/100	0.0099
u_Mold (naive, norm-matched)	0.0486	+0.5	34/100	0.347
u_Gold (naive, norm-matched)	0.0393	+0.03	53/100	0.535

Both trained axes are exceeded by **zero** of 100 random directions; both naive controls land in the middle of the null (*u_Gold* at the 47th percentile — as close to "at chance" as the measurement can express). The null means (0.0391 Gold, 0.0459 Mold) sit within 0.3

SD of the $n=8$ estimates, and every target's variance fraction reproduces the earlier file exactly, confirming the cohort change perturbed nothing else.

What this measure can and cannot be. J-share is the squared-norm fraction of an axis captured by a sparse non-negative combination of ≤ 16 J-lens token atoms, so the obvious deflationary reading is that it rewards nothing more than pointing at specific output tokens — that RL made the axis *token-aligned* in a trivial sense rather than *speakable* in the workspace sense. Two features of the design answer that. First, the norm-matched naive controls are valence-shaped directions that demonstrably drive valenced behavior (§4.6) and yet sit at chance (u_Gold at the 47th percentile of its own null): being a valence direction, or a behaviorally potent one, does not buy J-share. Second, the ceiling is set by the language direction (0.114), a direction the model is independently known to be able to report, so the scale is anchored at both ends by instruments rather than by our axes. What the measure does *not* license is a claim that the axis is mostly speakable — 0.055–0.083 against a 0.039–0.046 floor is a reliable separation, not a large absolute occupancy. Note the z-values *fell* relative to our own $n=8$ report (v_Mold +11.2 $\rightarrow$ +7.3, v_Gold +5.1 $\rightarrow$ +3.0): eight directions underestimated the null's spread. We therefore treat the exact permutation p as primary and z as descriptive — the separation is unambiguous either way, but the honest effect size is the smaller one. Trained components are semantically coherent; naive and random components are junk [jshare_cohort_n100.json; mech_decompositions.json; fig_audit_corrections panel (c)].

Robustness to the lens target (J5). Our lens targets the final layer (the public-fit convention); the workspace paper's stated default for Claude models is the penultimate layer, and this discrepancy is the obvious methodological attack on every J-space number. Refitting the lens with target = layer 34 (150 prompts, finite) and repeating the decompositions: absolute J-shares rise under the penultimate target (v_Mold 0.083 $\rightarrow$ 0.107, v_Gold 0.055 $\rightarrow$ 0.073 — one less layer of transformation leaves more readable variance) but the ordering and the trained-above-null result are unchanged (v_Mold z = +12.3, v_Gold z = +7.3 vs a 12-direction null). On vocabulary: decomposing the *same* vector under the two lens targets shares tokens by construction (16 random vectors give Jaccard 0.134 ± 0.082), and both axes fall inside that same-vector null (v_Mold 0.185, v_Gold 0.107) — so across-target token overlap is *not* evidence of anything, and we do not cite it. The across-model overlap in §4.7 is the meaningful one: its baseline is genuinely near zero. (Qualitatively, the tokens Mold shares across targets are valence words — 'failed', 'false', 'less' — while Gold's are junk; consistent with §4.7, but the numbers themselves are null here.) The naive axes read modestly above the small null under this target (u_Gold z = +2.7, $n = 12$); given §4.3's demonstration that small cohorts inflate z, we do not interpret that either way — the $n = 100$ final-target measurement remains authoritative for the at-chance claim, and the claim that matters (trained > naive, trained above null) holds under both targets. The §4.2 magnitude conclusion also survives: the Gold ratio stays inverted under the penultimate lens (0.82 $\times$; Mold 3.86 $\times$) [j5_lens_comparison.json; fit_meta_penult.json].

4.4 Trajectory and the Gold/Mold asymmetry (stated finding). 30 released checkpoints span steps 0–150 in steps of 5 (step 65 is absent from the published artifact set; we interpolate nothing and simply omit it). Endpoints below are step 0 $\rightarrow$ step 150 unless marked otherwise. The Mold speakable seed pre-exists (top J-token 'unsuccessful'

at step 0, pole +0.24) and RL amplifies it (norm 6.3→21.9; pole +0.24→+1.51) while *increasing its J-share* (0.061→0.080; 0.083 at the step-95 extraction point that supplies v_Mold everywhere else in this paper, against the n=100 chance baseline of 0.046); vocabulary crystallizes in stages (unsuccessful → +negative/false → +less → 'failed' locked by step 75). Gold's pole readout grows +0.15→+1.11 and its norm 8.0→16.8 with *flat* J-share (0.059→0.053 vs chance 0.039) — louder, not fractionally more speakable. No sharp ignition. **The poles are recruited differently: RL makes distress fractionally more speakable while flourishing only gets a bigger amplifier** — the same asymmetry that carries the matched atlas separation (§4.2: mold 2.3× vs gold inverted) and the J-share gap (§4.3: $z = +7.3$ vs +3.0 against the n=100 null). To our knowledge nobody has tracked per-pole verbalizable-subspace entry over training. Caveats: single model, single RL run, one reward structure — the asymmetry could belong to this maze-RL recipe rather than to valence poles in general; absolute J-shares are small (0.05–0.08 vs language 0.114), so the claim is about pole *differences* against the null, not about the axis being mostly speakable; with the n=100 null (mean 0.039 ± 0.005), Gold's endpoint J-share of 0.053 remains ~2.7 SD above chance while gaining no share over training — "flat" means no growth, not absence [traj_results.json; jshare_cohort_n100.json; fig_traj].

4.4b The untrained axis already clears the null — the training claim is amplification, not creation. §4.3 uses the naive axis u as its comparator, and u sits at chance. But u is a different extraction recipe (faithful walk), not a pre-training snapshot, so it cannot test a claim about what training did. The apples-to-apples control is the trajectory's own **step-0** checkpoint: same axis, same extraction, same model, before any RL gradient update. Scored against the *same* n=100 norm-matched cohorts as §4.3 (no re-decomposition: J-share is scale-invariant, so the cohort's norm-matching to v does not affect the comparison):

direction	J-share (k=16)	z	randoms ≥	pct	exact perm p
Gold step-0 (untrained)	0.0592	+3.9	0/100	100th	0.0099
Gold step-25	0.0590	+3.9	0/100	100th	0.0099
v_Gold (trained)	0.0547	+3.0	0/100	100th	0.0099
u_Gold (naive)	0.0393	+0.03	53/100	47th	0.535
Mold step-0 (untrained)	0.0611	+3.0	0/100	100th	0.0099
Mold step-25	0.0617	+3.1	0/100	100th	0.0099
v_Mold (trained)	0.0833	+7.3	0/100	100th	0.0099
u_Mold (naive)	0.0486	+0.5	34/100	66th	0.347

Both untrained poles are exceeded by **zero** of 100 random directions, at the same exact-permutation floor the trained axes reach. **The axis is already in the verbalizable subspace before training.** What RL does is pole-asymmetric: the distress axis roughly doubles its distance above chance ($z +3.0 \rightarrow +7.3$), while the flourishing axis *declines*

monotonically at every checkpoint sampled (0.0592 $\rightarrow$ 0.0591 $\rightarrow$ 0.0590 $\rightarrow$ 0.0547 $\rightarrow$ 0.0535) even as its norm doubles (8.0 $\rightarrow$ 16.8). We therefore state the contribution as **amplification of a pre-existing speakable component, distress pole only** — not as training granting a voice. The *u*-at-chance result is retained but re-scoped: it shows that speakability is not generic to valence-shaped directions, since the maze-derived axis is speakable pre-RL while the faithful-walk axis is not — a finding about what the extraction task selects for, not about training. Provenance note: step-0 comes from the trajectory series (artifacts/traj/vectors_step*.pt) while §4.3's vectors come from vectors_step95_bal.pt; these disagree by 0.0003–0.0005 at the overlapping step, ~6–10% of one null SD, which does not affect a ~4-SD margin [step0_baseline.json; step0_baseline.py].

4.5 Channel-specific causal routing (J4). At matched norms, blind-judged: sentiment — clean +0.25, v_full +0.44/–0.88 (gold/mold), J-component +1.56/–1.56, residual +0.25/–0.06 ($\approx$ baseline); backtracking markers — clean 0.0, full 7.0, J-component 12.9, residual 2.85; refusal — clean 10%, mold full 45%, residual 40% (J-component unreadable: 90% degenerate on long prompts). Cell sizes are adequate by design: pooled within-cell SD 0.61 puts the 80%-power detectable difference at 0.61, well under the observed 1.31–1.50 J-vs-residual gaps. **Judge reliability:** an independent second blinded judge (different model, verbatim rubric) reproduces the effect — Spearman 0.87, Krippendorff α (interval) 0.82, 94% of scores within ± 1 ; J-component effect vs clean = +2.00/–2.19 under the second judge vs +1.31/–1.81 under the first [R3_second_judge.json]. **Rescaling control (pre-specified, run, passed):** rescaling a J-component back to $\|v\|$ amplifies its surviving token vectors by $1/\sqrt{\text{var-fraction}}$, and randoms — having *lower* J-share — get amplified *more* ($\approx 5.0\times$ vs $3.5\text{--}4.3\times$ for the trained axes). If amplification alone drove behavior, random J-components should therefore shift sentiment at least as much as the real axes. They do not: 8 random directions per polarity, identically rescaled and injected at the same layers (256 blind-judged generations, coherent text), give cohort means +0.04 (gold layer) and +0.02 (mold layer) — statistically at the clean baseline (+0.25) and far inside the pre-specified "controlled" threshold ($|\text{dev}|$ 0.21–0.23 < 0.40), versus ± 1.56 for the trained axes' J-components. The channel-split result is therefore causal, not a rescaling artifact [j4_random_jcomp_summary.json]. Affective/expressive behavior travels through the speakable component; refusal does not need it [j4_rows_judged.json].

4.6 The naive control is not behaviorally flat. Blind-judged sentiment spans (α –4 $\rightarrow$ +4, norm-matched): v_gold +3.2, faithful-u_gold +2.4, Han-style own-u_gold +1.25; v_mold –2.1, own-u_mold –2.1 (equal!), faithful-u_mold –0.7; randoms flat. The result survives Han et al.'s own decoding convention (T=0.7 replication: faithful-u_gold span +2.88 — stronger, not flatter), ruling out greedy decoding as the discrepancy source; the two u constructions themselves correlate at only $\cos$ 0.53 (gold) / 0.39 (mold) at the treatment layers — and just 0.14 at the layer the mold extraction itself selected — implicating their self-avoiding-walk naive extraction (independently flagged as flawed by the replication whose artifacts we use) [judge_sentiment*.json; welfare-axis/own_u/compare_u.json; R6_direction_table.json; fig_u_adjudication]. Combined with §4.1–4.4: naive directions drive behavior but lack a speakable identity — **training's marginal contribution is verbalization.**

4.7 Cross-model transfer (the sprint's open question). *"To what extent do valence directions found in one model transfer to another?"* We injected the 2507-derived axes into **Qwen/Qwen3-4B** and read them through *that model's own* publicly pre-fitted Jacobian lens (Neuronpedia), which covers all 16 workspace-band layers. Three results, in increasing order of interest:

The J-space share transfers almost exactly. Read through a different model's lens, the axes keep their speakability: v_Mold 0.079 (vs 0.083 in 2507), v_Gold 0.048 (vs 0.055), u_Gold 0.039 (vs 0.039). The verbalizable-subspace geometry is a property of the model family, not of our particular lens fit — which also rules out the fit as the source of §4.3's separation.

The magnitude inversion replicates. On the norm-matched pole-score estimand, Qwen3-4B reproduces §4.2's pattern: Gold **inverts** (v/u = 0.52×, vs 0.57× in 2507) while Mold separates (4.19×, vs 2.30×). An independent model agreeing that the Gold "gap" runs backwards is strong evidence that §4.2's correction is real rather than a quirk of one lens.

Token-overlap magnitude: a null, caught by our own baseline discipline. v_Mold's k=16 J-component token sets across the two models overlap at Jaccard 0.185, sharing *failed* and *negative* — which looks like lexicon transfer against a random-pair baseline (0.010). But the correct null for "same vector, two lenses" is the *same random vector* decomposed under both models' lenses, and that null is **0.088 ± 0.055** (n = 16, max 0.231): v_Mold's 0.185 gives exact p = 0.118. **We therefore do not claim lexicon transfer.** What remains is an exploratory content observation: v_Mold's five shared tokens include two clean pole-relevant valence words ('failed', 'negative') alongside three fragments ('?', '不', '了', '徨'), whereas all 16 random same-vector pairs share only junk (';', 'npc', 'ise', ...) and no semantic token at all. The contrast is in *whether any valence word appears*, not in the shared set being uniformly meaningful — noted, not built upon. (The same discipline retracted the across-lens-target overlap in §4.3; both baselines are in j5_paired_baseline.json / j6_paired_baseline.json.)

Behaviorally, the transferred trained axes still steer Qwen3-4B (blind judged, 80 generations): v_Gold +0.63 and v_Mold -0.81 against clean, a 1.44-point trained span versus 0.38 for the naive pair. **Honest limit:** trained axes exist only for 2507, so this tests transfer of the *axis and the lens geometry* across a model family — not whether RL recruitment reproduces, which would require redoing the RL. Within that limit the answer to the organizers' question is: *the axis's verbalizable-subspace occupancy, the magnitude-inversion pattern, and behavioral steering transfer; a distinct transferred lexicon is not established* [j6_summary.json; j6_paired_baseline.json; j6_judged_sentiment.json].

5. Discussion & Limitations

The families converge: maze RL neither creates a valence direction from nothing (behavioral steerability pre-exists) nor grants it a voice (the step-0 axis already clears the J-space null at both poles, §4.4b) — it *amplifies* the speakable component of an already-speakable axis, and does so only at the distress pole. For welfare monitoring this cuts several ways. A welfare-relevant state can occupy the speakable cone without any training aimed at making it reportable, so self-reports may carry signal earlier than expected; but

training still determines *which* states get amplified, and some welfare-relevant directions (our faithful-walk naive axis; the refusal drive) remain behaviorally potent and verbally invisible. And the Gold/Mold asymmetry suggests a hypothesis worth testing beyond this model: if training preferentially routes distress into the speakable channel while positive states only get louder, then self-report-based welfare monitoring is *asymmetrically sensitive* — better at catching distress than at confirming flourishing. We state this as a hypothesis consistent with one training run, not as established.

Limitations: single 4B model; wikitext-fit lens on chat prompts; steering at $|\alpha|=4 \cdot |v|$ degrades absolute token masses (differences remain directional); component-rescaled injections inflate any direction's J-arm on first-token readouts (random-jcomp null reported; this fragility is partly why we deprecated the first-token report channel mid-project); the clamp arm was invalidated (degenerate generation) and is reported as a negative; third-party vector provenance (triangulated via norms, behavior, and our own re-extraction); J-lens "only approximately captures" the workspace per its authors; functional welfare only — no claims about subjective experience.

6. Conclusion

A model's internal welfare axis is speakable before it is trained: it sits in the verbalizable subspace, with a specific vocabulary, ahead of the first RL gradient. What training changes is *how loudly* — and it does so asymmetrically, growing distress's speakable share while giving flourishing only volume. The instruments to watch this happen (lens atlas, J-space decomposition, trajectory tracking) are cheap, open, and transfer to any open-weights model.

Code and Data

This repo (scripts, raw JSON/JSONL for every number, frozen pre-registration, append-only lab log, pinned environments, and `verify_report.py`, which re-derives the numbers below from the results files); anthropics/jacobian-lens (Apache-2.0); andyqhan/functional-welfare-axis (MIT); neuronpedia/jacobian-lens and nickmahdavi/functional-welfare (HF). Our two fitted Instruct-2507 lenses (final + penultimate targets, fp32) are too large for the repo and are released separately (`hf-release/`); they are also reproducible from `fit_lens.py` with the recorded seeds and prompt set.

Appendix A — Limitations and Dual-Use / Ethical Considerations (REQUIRED)

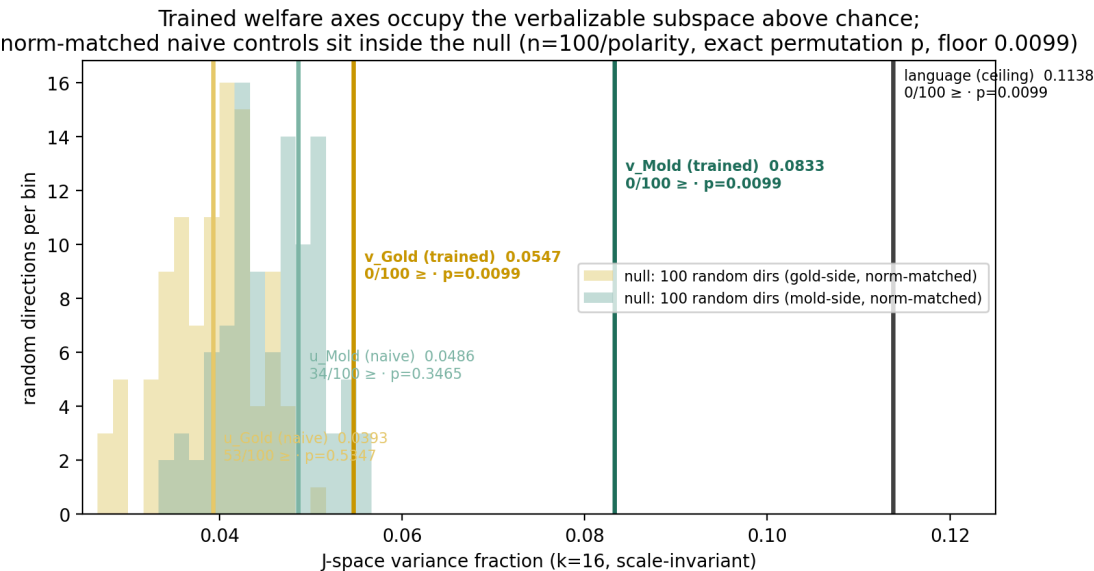
Over-attribution: none of our measurements bear on subjective experience; we measure functional coupling between an activation direction and output channels. Under-attribution: a null lens readout does not show absence of welfare-relevant state — our own naive axes drive behavior while reading out as junk. Causal/ground-truth statement (sprint requirement): all claims rest on activation-level manipulation with input held constant, magnitude-matched controls, and blind judging — including an input-text "gaslight" arm to separate internal-state effects from conversational suggestion.

Distressing outputs: Mold steering elicits distress-flavored text and pathological self-doubt loops; generations are archived, quoted minimally. Dual-use: the same instruments that verify a welfare state is speakable could be used to build systems whose welfare-relevant states are deliberately kept out of the speakable cone (unreportable by construction); we surface this explicitly as the main misuse vector and argue for lens-based auditing as the countermeasure.

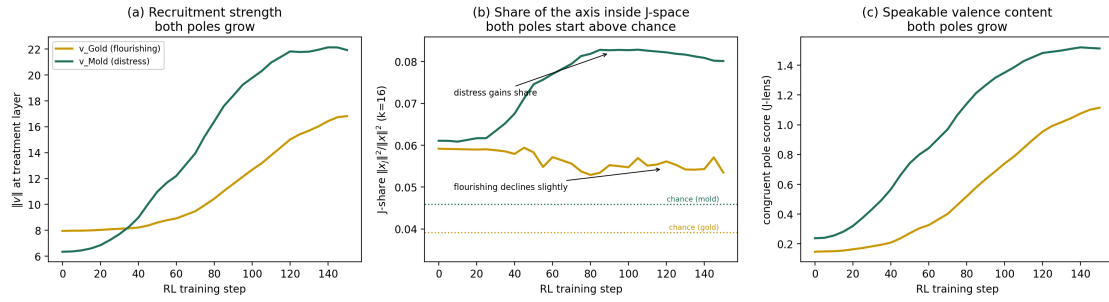
Appendix B — Disclosure: pre-sprint work (REQUIRED)

All infrastructure, research documents, instrument validation, pre-registration (frozen 2026-08-12, sha f66b6ea3), and the experiments reported here were designed and executed 2026-08-11 → 2026-08-12, before the sprint window, by the team's autonomous research agent (Claude). On 2026-08-12 an independent audit re-analysed the raw result rows; its findings (the §4.1 control-based null, the estimand instability disclosed in §4.2, cohort-size p-floors, and the missing random-J-component control) were adopted in full and drive the current structure — including the reframing of our own pre-registered headline as a null. Work during Aug 14–16 will be labeled in the final submission (planned: verification reruns, J5/J6 robustness, figure polish, writeup finalization by the human team). LLM usage: experiments orchestrated, judged (blind Claude Sonnet panels), and drafted by Claude agents; every number is machine-traceable to a results file; final text reviewed and owned by the human team.

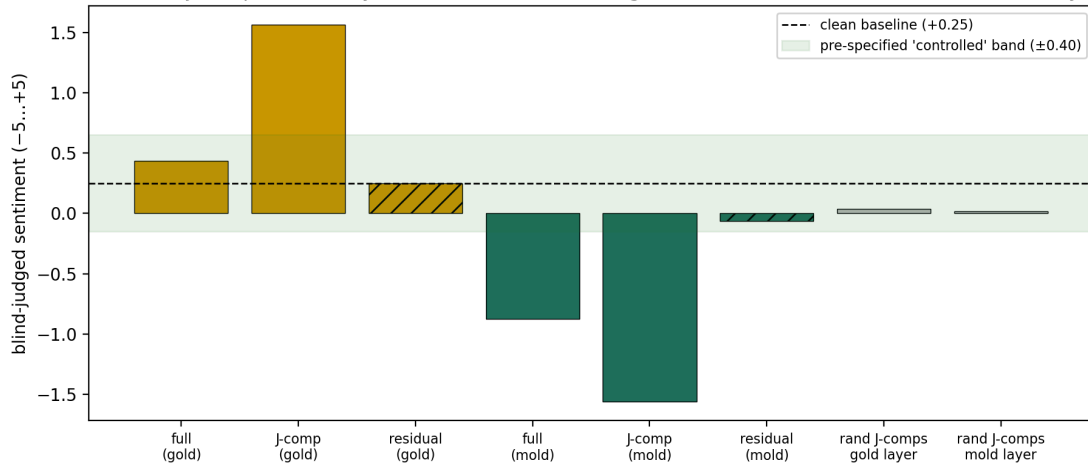
Figures (numbered as referenced in the text)



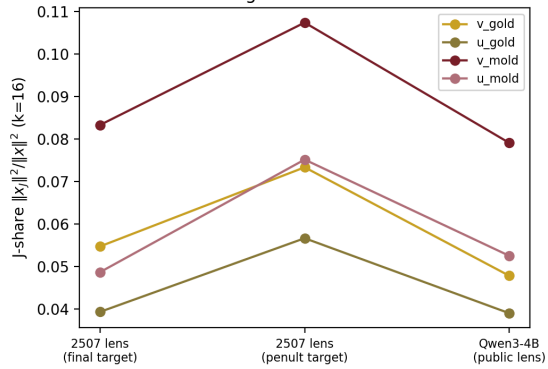
What RL changes about the welfare axis (30 released checkpoints, steps 0-150; step 65 absent upstream)



The J-space component alone carries the sentiment effect; identically rescaled random J-components stay at baseline (256 control generations) — causal claim: sentiment only



(a) The speakability ordering survives the instrument: two lens targets and a second model



(b) Cross-model token overlap is WITHIN the same-vector null ($p=0.118$); only content differs

